

4. Social Media location & Search Engine Analysis

Q.1) What is location analytics? Explain its significance in context of social media analytics.

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- Location analytics, also known as spatial analysis or geo-analytics is concerned with mapping, visualizing and mining the location of people, data and other resources.
- GPS-enabled mobile devices make it possible to provide location-based services, products and information.
- Location analytics can be used in various sectors and can be instrumental in understanding cities and human behavior in space and time.
- By adding location to your organization's analytics, you can enhance the context of your decision-making and gain better insights.
- Benefits of Location Analytics

1) Hyper-local intelligence

- ⇒ With location analytics, you can automatically transform your data into location-based insights using beautiful location-aware visualizations of GIS conversion.
- Communicate with everyone : business users, analysts, scientists, developers.

2) Real-world context

- ⇒ Unlike many other ~~visualize~~ information visualizations, maps connect data to the real-world and clearly show how location relate to other data characteristics.
- This context enriches the insights your analytics team and end users bring to your visualizations.

3) Actionable Insights

- ⇒ Location analysis enables users to instantly create insightful geographic analyses, without having to learn other GIS tools.
- It continues to provide the depth of analysis needed for predictive analytics and other location-based optimizations.

4) Targeted Marketing

- ⇒ Location data helps brands deliver geo-targeted ads or campaigns to specific audiences based on where they are.

For e.g.,

A coffee brand targeting users near its stores who mention "cold weather" or "morning".

B) Understanding of User behavior

⇒ By analyzing where users post from, businesses can identify regional preferences, user interests or local trends.

• For e.g.,

Analyzing tweets during a festival to see which cities are most engaged.

• Categories of Location Analytics:

- 1) Business data-driven
- 2) Social media data-driven

Q.2 Note on : Sources of location data

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1) Postal Address

⇒ Most business analytics applications rely on address information their customers, including city names, locality names and postal or zip codes.

2) GPS-based

⇒ GPS is a satellite-based navigation system that can be used to find exact location of people and resources.

- Mobile analytics mostly rely on GPS-based location data.
- GPS-based location analytics can provide us the most accurate location for social media users.
- Used in apps like Google Maps, ride-sharing.

3) IP-based

⇒ Public IP can be used to determine the location of Internet users.

- A public-IP address is an exclusive numerical address assigned to a device connected to the Internet.
- Different regions in the world are assigned a specific block of public IP addresses; hence it can be used to mine approximate geo-location of Internet users.
- Less precise than GPS but useful for region-level insights.

4) Image Metadata

- ⇒ Photos taken with smartphones or digital cameras often contain EXIF metadata, which includes GPS coordinates.
- Useful in platforms like Instagram, Flickr for geospatial analysis.

5) App and device data

- ⇒ Mobile apps collect data for navigation, delivery or personalized services.
- Often obtained with user permission.
- Example: Food delivery apps, weather apps.

Q.3 Discuss business data-driven and social media data-driven location analytics.

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- Categories of Location Analytics

- 1) Business Data-driven location analytics
- 2) Social Media Data-driven location analytics

1) Business Data-driven Location Analytics

⇒ Business data-driven location analysis deals with mapping, visualizing and mining location data to reveal patterns, trends and relationships hidden in tabular business data.

- By capitalizing on data stored in business databases, location analytics can provide location-specific information, products and services based on where customers are.
- Using the location of customers, for instance, it is possible to recommend the nearest convenience store, coffee shop, taxi or even probable social relations.
- Or it can be used for any other business decisions, such as what is the best potential new site for a business warehouse.
- Applications

- 1) Powerful Intelligence
- 2) Geo-enrichment
- 3) Collaboration and sharing

i) Powerful Intelligence

- ⇒ Simple maps can be useful, but they are limited in providing insightful details.
- Using sophisticated mapping techniques, such as clustering, heat mapping, data aggregation, and color-coded mapping, can generate powerful business intelligence.

ii) Geo-enrichment

- ⇒ Simple data maps can be enriched with customer data, including demographic, consumer spending, lifestyle and location.
- This can help to answer questions such as where's a loyal company's loyal customers spend most of their time.

iii) Collaboration and Sharing

- ⇒ Maps are easy to understand and are useful for communication and collaboration within an organization.
- Location analytics can be used to map business data for collaboration and information sharing with customers.
 - Business data - Google Fusion Tables, allows for the creation of maps, tables and charts.

2) Social Media Data-Driven Location Analytics

⇒ Social media data-driven analytics relies on social media location data to mine and map location of social media users, content and data.

- Social media location information comes mainly from GPS and IP.

- Applications :

i) Recommendation purposes :

⇒ organizations can harvest location data to recommend products, services and social events to potential customers in real-time.

- For e.g.,

Trinder recommends potential social relationships based on location of users.

ii) Customer segmentation

⇒ Social media location data can be used to segment customers based on their geographic location.

- Tweepismap can be used to geo-locate your Twitter followers by country, state or city.

iii) Advertisement

⇒ Location-based advertisement allows targeted marketing and promotions campaign mostly delivered through mobile devices to reach specific target audiences.

4) Alerts

⇒ Location data can be used to send and receive alerts and notifications, such as sales and promotion alerts, traffic congestion alerts, speed limit warnings and storm warnings.

5) Search and Rescue

⇒ Location data is vital in search and rescue operations.

Q.4 List all the location analytics tools and also explain working of every tool.

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1) Google Fusion Tables

⇒ Google Fusion Tables is a web service for geo-tagging, storing, sharing, querying and visualizing tabular business data on Google Maps.

- Allowed users to upload CSVs or spreadsheets, automatically detect geographic fields, and plot data on a map.

2) Agos

⇒ Agos is a platform geo-tagging and reporting that helps communities address climate change and disaster risk reduction.

3) Google Analytics

⇒ Google analytics helps you understand and improve certain goals, including which pages a visitor click on, how long they stay on certain page and what actions they take.

4) Tweepmap

⇒ Tweepmap is a tool that maps twitter followers based on their location.

- By analyzing the location data from twitter profiles, it visualizes the geographic distribution of followers on map.

- It also provides demographics, engagement metrics and audience insights to help user understand and target their social media strategy effectively.

5) Trendsmap

- ⇒ Trendsmap is a real-time visualization tool that displays trending topics on Twitter geographically.
- It fetches trending hashtag and keywords from Twitter and displays them on a global or regional map, allowing users to see what topics are trending in specific locations in real-time.
- It is widely used in journalism, marketing and social media analysis.

Q.5 Explain two main categories of search engine analytics.

- ⇒
- Search engines are the gateways to social media and help users search for and find information.
 - To be precise, a search engine is an Internet service or software designed to search information on the web that corresponds to a request specified by the user.
 - Considering that there are billions of websites over the web, search engines play a crucial role in helping us find the right information in a limited amount of time.

Search engine analytics

- 1) Search Engine Optimization (SEO)
- 2) Search engine trend analysis

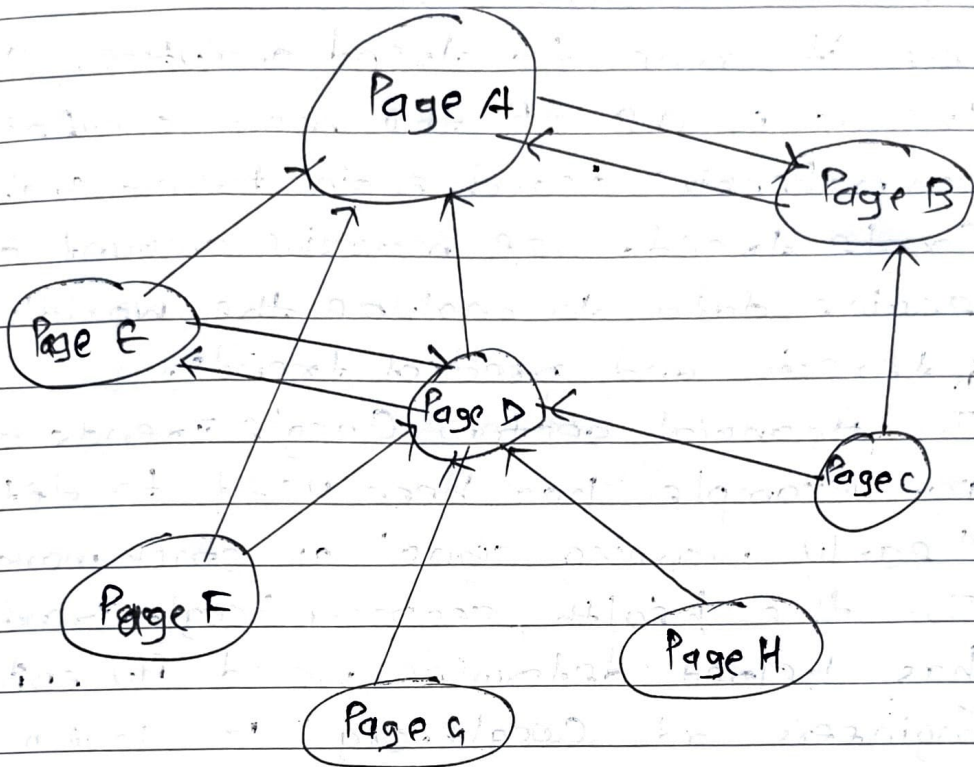
1) SEO

⇒ Search engine optimization (SEO) is a technique used to improve a website's ranking in a search result page engine result page (SERP).

- A SERP is the list of the results returned by a search engine in response to user's query.
- SERPs generally have two types of results: organic and non-organic search results.

- Organic results appear mainly because of their relevance to the user's query.
- Non-organic search results include paid advertisement.
- It is important for the websites to appear ~~to~~ at top in the SERP, as users pay closer attention to the top results on search engines.
- The ranking becomes more crucial when the website is commercial by nature, that is selling products or services.
- High ranking on SERPs can mean more Internet traffic to a website, which in some cases converts to more paying clients and higher return on investment.
- However, the most important factor that determines SERP ranking is PageRank.
- PageRank is a mechanism used by Google search engines to rank websites' SERPs.
- The website that rank higher are displayed on the top of the search result page.
- Google's PageRank algorithm predominantly relies on the quality of incoming hyperlinks to rank websites.
- A website, for example, with in-links from a famous website will appear on the top of the SERP if compared with a website with no quality in-links or many low-quality in-links.

- Consider following figures, where node represents web pages and lines represents in-links and out-links.



- The pagerank algorithm will place page B higher on the SERP, even though there are fewer in-links to B when compared to D.
- The reason for this ranking is that in-links to website B are from an important website; that is A.

2) Search Engine Trend Analytics

⇒ Search engine trend analytics deals with analyzing and understanding the keywords people use in a search engine.

- Search engine data are gateways into the minds of customers.

- Through search engines, customers search for what they want, thus search trend analysis can provide valuable information to the social marketers.
- When it comes to trend analytics, Google Trends is one of the most convenient and comprehensive search engine trend analysis tools.
- Google Trends use massive amount of search engine data to analyze the world's interests and predict trends.
- In financial sector, Google Trends data, for example, has been used to detect "early warning signs" of stock market moves.
- In the health sector, Google Trends data has helped determine world flu epidemics.
- Engineers at Google.org, for instance, using Google Trends data, found a strong correlation among the searches for flu-related topics and the number of actual flu cases circulating in different countries and regions around the world.

Q.6 What is purpose of SEO?

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1) Increase website visibility

⇒ Help your site appear on the first page of search results, where users are most likely to click

- 75% of users never scroll past the first page on Google.

2) Drive Organic Traffic

⇒ Attract more unpaid visitors to your website.

- Organic traffic is more sustainable and cost-effective than paid advertising.

3) Reach targeted Audiences

⇒ Connect with people who are actively searching for your products, services or content.

- Leads generated from SEO tend to have higher conversion rates because the intent already exists.

4) Enhance User Experience

⇒ Make websites faster, easier to navigate and more accessible.

- Search engines reward websites that offer a better user experiences with higher rankings.

Q.7 What is SEO? What are different methods to do it?

- ⇒
- Search Engine Optimization is a technique used to improve a website's ranking in Search Engine Page Result (SERP).
 - A SERP is a list of results returned by a search engine in response to the user's query.

- Types of SEO

- 1) On-Page SEO
- 2) Off-Page SEO
- 3) Technical SEO
- 4) Local SEO
- 5) E-commerce SEO
- 6) Social SEO

1) On-Page SEO

⇒ The first and foremost type of SEO optimization is On-Page SEO.

- It refers to the optimization of individual web pages to improve their ranking in SERP.
- Meta tags: Optimizing meta titles and descriptions for better indexing.
- Keyword Optimization: Using relevant keywords in titles, headings and content.
- URL Structure: Creating clean, readable URLs with keywords.

2) OFF-Page SEO

⇒ OFF-page SEO refers to the optimization of factors outside of a website to improve its ranking in SERPs.

- Backlink building: Getting links from other high-authority websites.
- Social media sharing: Promoting content through social platforms.
- Guest blogging: Writing content on another websites with links back to your site.

3) Technical SEO

⇒ It refers to the optimization of the technical aspects of a website to improve its ranking in SERPs.

- Website speed: Ensures the site loads quickly.
- Mobile Friendliness: Making sure that site works well ~~with~~ on mobile devices.

4) Local SEO

⇒ In this, set of strategies and techniques used to optimize a website and its content to improve its visibility and ranking in local search results.

- Local keywords: Using keywords with city or region names.
- Google My business optimization: Updating business details ~~with~~ and posts.

5) E-commerce SEO

⇒ E-commerce SEO involves optimizing online stores to improve their visibility in search engine results and attract potential customers.

- Keyword Research: Identifying and targeting relevant keywords related to the products offered on e-commerce site.
- User-friendly URLs: Creating clean and user-friendly URLs that include relevant keywords and accurately describes the content.

6) Social SEO

⇒ Social SEO is a strategy that involves optimizing your website and social media presence to improve your search engine rankings.

- It combines the principles of SEO with social media marketing to help your website appear higher in search results and attract more visitors from social media platform.

Q.3 Discuss privacy concerns related to location analytics?

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- Social media privacy refers to personal and confidential information that other people can learn about you through your account.
- This information may be shared intentionally or unknowingly.

1) Hacking and account takeovers

- ⇒ Many people unknowingly post personal information that can give hackers clues about passwords and security questions.
- For example, posts about your hometown, pets, elementary schools or extended family.
 - Scammers use this information to try to force their way into your account or use social engineering attacks to trick you into entering your password.

2) Social media Phishing scam

- ⇒ If your social media account isn't set to private, anyone can message you.
- This includes scammer trying to get you to click on malicious links.
 - Last year, 12% of his total clicks on fake phishing websites came from social media.
 - Scammers also use social media to carry out love and investment fraud schemes.

3) Shared location data used by stalkers and predators

⇒ Many social media websites include location data by default.

- This data can be used by stalkers, scammers and even thieves to track your movements.

4) Third-party apps that can access your other accounts

⇒ Many people use social media logins (such as "login with Facebook").

- However, as convenient as these services are, they may share your personal information with companies and apps that don't have the best digital security.

5) Cybersecurity and Online harassment

⇒ For children, teens and even adults, social media can be a source of bullying and emotional and psychological aggression.

- A public account gives perpetrators of cyberbullying easy access to attack you with messages and malicious posts, or access your personal information.

Q.6 What is search engine? Discuss the different types of search engines.

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- A search engine is a software system designed to retrieve information from the World Wide Web or database based on specific user queries.
- These queries typically consists of keywords, or phrases entered by users and the search engine returns a list of relevant web pages, documents or other types of content.
- Types :

→ 1. Crawler-Based

- i) Web crawling
- ii) Indexing
- iii) Searching

→ 2. Directories

→ 3. Metasearch engines

- i) Local search engines
- ii) Global search engines

1) Crawler-based

- ⇒ As the name suggests, crawler-based search engines create their databases or list automatically, without any human intervention.
- For eg, crawler-based search engines are Google.com, Bing.com

i) Web Crawling: Search engine start by collecting and storing information about web pages. This mechanism is termed as web crawling.

ii) Indexing: Indexing helps classify a website correctly for searching purposes. The data crawled or extracted is then indexed and stored in database for quick access. Common indexing techniques include storing meta data and keywords related to a website.

iii) Searching

⇒ Searching is the final step in search engine operation. When a user requests specific information by entering keyword in a search engine, the search engine queries the index and provides a list of the most relevant web pages by matching it with the indexed keywords.

2) Directories

⇒ The listings in directories are manually compiled and created by human editors.

People who want to be listed in a directory submit an address, title and brief description of their website, which is then reviewed by the editor and included in it.

Some good examples of human-created directories are Yahoo directory, Open directory and LookSmart.

3) Metasearch Engine

→ Metasearch engine compiles and displays results from other search engines.

Examples of metasearch engines include MetaCrawler, Mamma and Dogpile.

i) Local Search engines: A search engine is local in that the sense that it is embedded within a website and only indexes and searches the content of that website. For e.g., Amazon's cloud search.

ii) Global search engines: Global search engines are used to search for content on the web.

For e.g., Google.com, Bing.com.